

21.1 Basic configuration

The Timer 0, 1, 2, and 3 peripherals are configured using the following registers:

1. Power: In the PCONP register ([Table 46](#)), set bits PCTIM0/1/2/3.
Remark: On reset, Timer0/1 are enabled (PCTIM0/1 = 1), and Timer2/3 are disabled (PCTIM2/3 = 0).
2. Peripheral clock: In the PCLKSEL0 register ([Table 40](#)), select PCLK_TIMER0/1; in the PCLKSEL1 register ([Table 41](#)), select PCLK_TIMER2/3.
3. Pins: Select timer pins through the PINSEL registers. Select the pin modes for the port pins with timer functions through the PINMODE registers ([Section 8.5](#)).
4. Interrupts: See register T0/1/2/3MCR ([Table 430](#)) and T0/1/2/3CCR ([Table 431](#)) for match and capture events. Interrupts are enabled in the NVIC using the appropriate Interrupt Set Enable register.
5. DMA: Up to two match conditions can be used to generate timed DMA requests, see [Table 544](#).

21.2 Features

Remark: The four Timer/Counters are identical except for the peripheral base address. A minimum of two Capture inputs and two Match outputs are pinned out for all four timers, with a choice of multiple pins for each. Timer 2 brings out all four Match outputs.

- A 32-bit Timer/Counter with a programmable 32-bit Prescaler.
- Counter or Timer operation
- Up to two 32-bit capture channels per timer, that can take a snapshot of the timer value when an input signal transitions. A capture event may also optionally generate an interrupt.
- Four 32-bit match registers that allow:
 - Continuous operation with optional interrupt generation on match.
 - Stop timer on match with optional interrupt generation.
 - Reset timer on match with optional interrupt generation.
- Up to four external outputs corresponding to match registers, with the following capabilities:
 - Set low on match.
 - Set high on match.
 - Toggle on match.
 - Do nothing on match.

21.3 Applications

- Interval Timer for counting internal events.
- Pulse Width Demodulator via Capture inputs.
- Free running timer.

21.4 Description

The Timer/Counter is designed to count cycles of the peripheral clock (PCLK) or an externally-supplied clock, and can optionally generate interrupts or perform other actions at specified timer values, based on four match registers. It also includes four capture inputs to trap the timer value when an input signal transitions, optionally generating an interrupt.

21.5 Pin description

[Table 425](#) gives a brief summary of each of the Timer/Counter related pins.

Table 425. Timer/Counter pin description

Pin	Type	Description
CAP0[1:0] CAP1[1:0] CAP2[1:0] CAP3[1:0]	Input	Capture Signals- A transition on a capture pin can be configured to load one of the Capture Registers with the value in the Timer Counter and optionally generate an interrupt. Capture functionality can be selected from a number of pins. When more than one pin is selected for a Capture input on a single TIMER0/1 channel, the pin with the lowest Port number is used Timer/Counter block can select a capture signal as a clock source instead of the PCLK derived clock. For more details see Section 21.6.3 .
MAT0[1:0] MAT1[1:0] MAT2[3:0] MAT3[1:0]	Output	External Match Output - When a match register (MR3:0) equals the timer counter (TC) this output can either toggle, go low, go high, or do nothing. The External Match Register (EMR) controls the functionality of this output. Match Output functionality can be selected on a number of pins in parallel.

21.5.1 Multiple CAP and MAT pins

Software can select from multiple pins for the CAP or MAT functions in the Pin Select registers, which are described in [Section 8.5](#). When more than one pin is selected for a MAT output, all such pins are driven identically. When more than one pin is selected for a CAP input, the pin with the lowest Port number is used. Note that match conditions may be used internally without the use of a device pin.

Effective processing of the externally supplied clock to the counter has some limitations. Since two successive rising edges of the PCLK clock are used to identify only one edge on the CAP selected input, the frequency of the CAP input can not exceed one quarter of the PCLK clock. Consequently, duration of the high/low levels on the same CAP input in this case cannot be shorter than $1/(2 \times \text{PCLK})$.

21.6 Register description

Each Timer/Counter contains the registers shown in [Table 426](#) ("Reset Value" refers to the data stored in used bits only; it does not include reserved bits content). More detailed descriptions follow.

Table 426. TIMER/COUNTER0-3 register map

Generic Name	Description	Access	Reset Value ^[1]	TIMERN Register/ Name & Address
IR	Interrupt Register. The IR can be written to clear interrupts. The IR can be read to identify which of eight possible interrupt sources are pending.	R/W	0	T0IR - 0x4000 4000 T1IR - 0x4000 8000 T2IR - 0x4009 0000 T3IR - 0x4009 4000
TCR	Timer Control Register. The TCR is used to control the Timer Counter functions. The Timer Counter can be disabled or reset through the TCR.	R/W	0	T0TCR - 0x4000 4004 T1TCR - 0x4000 8004 T2TCR - 0x4009 0004 T3TCR - 0x4009 4004
TC	Timer Counter. The 32-bit TC is incremented every PR+1 cycles of PCLK. The TC is controlled through the TCR.	R/W	0	T0TC - 0x4000 4008 T1TC - 0x4000 8008 T2TC - 0x4009 0008 T3TC - 0x4009 4008
PR	Prescale Register. When the Prescale Counter (below) is equal to this value, the next clock increments the TC and clears the PC.	R/W	0	T0PR - 0x4000 400C T1PR - 0x4000 800C T2PR - 0x4009 000C T3PR - 0x4009 400C
PC	Prescale Counter. The 32-bit PC is a counter which is incremented to the value stored in PR. When the value in PR is reached, the TC is incremented and the PC is cleared. The PC is observable and controllable through the bus interface.	R/W	0	T0PC - 0x4000 4010 T1PC - 0x4000 8010 T2PC - 0x4009 0010 T3PC - 0x4009 4010
MCR	Match Control Register. The MCR is used to control if an interrupt is generated and if the TC is reset when a Match occurs.	R/W	0	T0MCR - 0x4000 4014 T1MCR - 0x4000 8014 T2MCR - 0x4009 0014 T3MCR - 0x4009 4014
MR0	Match Register 0. MR0 can be enabled through the MCR to reset the TC, stop both the TC and PC, and/or generate an interrupt every time MR0 matches the TC.	R/W	0	T0MR0 - 0x4000 4018 T1MR0 - 0x4000 8018 T2MR0 - 0x4009 0018 T3MR0 - 0x4009 4018
MR1	Match Register 1. See MR0 description.	R/W	0	T0MR1 - 0x4000 401C T1MR1 - 0x4000 801C T2MR1 - 0x4009 001C T3MR1 - 0x4009 401C
MR2	Match Register 2. See MR0 description.	R/W	0	T0MR2 - 0x4000 4020 T1MR2 - 0x4000 8020 T2MR2 - 0x4009 0020 T3MR2 - 0x4009 4020
MR3	Match Register 3. See MR0 description.	R/W	0	T0MR3 - 0x4000 4024 T1MR3 - 0x4000 8024 T2MR3 - 0x4009 0024 T3MR3 - 0x4009 4024
CCR	Capture Control Register. The CCR controls which edges of the capture inputs are used to load the Capture Registers and whether or not an interrupt is generated when a capture takes place.	R/W	0	T0CCR - 0x4000 4028 T1CCR - 0x4000 8028 T2CCR - 0x4009 0028 T3CCR - 0x4009 4028

Table 426. TIMER/COUNTER0-3 register map ...continued

Generic Name	Description	Access	Reset Value ^[1]	TIMERN Register/ Name & Address
CR0	Capture Register 0. CR0 is loaded with the value of TC when there is an event on the CAPn.0(CAP0.0 or CAP1.0 respectively) input.	RO	0	T0CR0 - 0x4000 402C T1CR0 - 0x4000 802C T2CR0 - 0x4009 002C T3CR0 - 0x4009 402C
CR1	Capture Register 1. See CR0 description.	RO	0	T0CR1 - 0x4000 4030 T1CR1 - 0x4000 8030 T2CR1 - 0x4009 0030 T3CR1 - 0x4009 4030
EMR	External Match Register. The EMR controls the external match pins MATn.0-3 (MAT0.0-3 and MAT1.0-3 respectively).	R/W	0	T0EMR - 0x4000 403C T1EMR - 0x4000 803C T2EMR - 0x4009 003C T3EMR - 0x4009 403C
CTCR	Count Control Register. The CTCR selects between Timer and Counter mode, and in Counter mode selects the signal and edge(s) for counting.	R/W	0	T0CTCR - 0x4000 4070 T1CTCR - 0x4000 8070 T2CTCR - 0x4009 0070 T3CTCR - 0x4009 4070

[1] Reset Value reflects the data stored in used bits only. It does not include reserved bits content.

21.6.1 Interrupt Register (T[0/1/2/3]IR - 0x4000 4000, 0x4000 8000, 0x4009 0000, 0x4009 4000)

The Interrupt Register consists of 4 bits for the match interrupts and 4 bits for the capture interrupts. If an interrupt is generated then the corresponding bit in the IR will be high. Otherwise, the bit will be low. Writing a logic one to the corresponding IR bit will reset the interrupt. Writing a zero has no effect. The act of clearing an interrupt for a timer match also clears any corresponding DMA request.

Table 427. Interrupt Register (T[0/1/2/3]IR - addresses 0x4000 4000, 0x4000 8000, 0x4009 0000, 0x4009 4000) bit description

Bit	Symbol	Description	Reset Value
0	MR0 Interrupt	Interrupt flag for match channel 0.	0
1	MR1 Interrupt	Interrupt flag for match channel 1.	0
2	MR2 Interrupt	Interrupt flag for match channel 2.	0
3	MR3 Interrupt	Interrupt flag for match channel 3.	0
4	CR0 Interrupt	Interrupt flag for capture channel 0 event.	0
5	CR1 Interrupt	Interrupt flag for capture channel 1 event.	0
31:6	-	Reserved	-

21.6.2 Timer Control Register (T[0/1/2/3]CR - 0x4000 4004, 0x4000 8004, 0x4009 0004, 0x4009 4004)

The Timer Control Register (TCR) is used to control the operation of the Timer/Counter.

Table 428. Timer Control Register (TCR, TIMERN: TnTCR - addresses 0x4000 4004, 0x4000 8004, 0x4009 0004, 0x4009 4004) bit description

Bit	Symbol	Description	Reset Value
0	Counter Enable	When one, the Timer Counter and Prescale Counter are enabled for counting. When zero, the counters are disabled.	0
1	Counter Reset	When one, the Timer Counter and the Prescale Counter are synchronously reset on the next positive edge of PCLK. The counters remain reset until TCR[1] is returned to zero.	0
31:2	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

21.6.3 Count Control Register (T[0/1/2/3]CTCR - 0x4000 4070, 0x4000 8070, 0x4009 0070, 0x4009 4070)

The Count Control Register (CTCR) is used to select between Timer and Counter mode, and in Counter mode to select the pin and edge(s) for counting.

When Counter Mode is chosen as a mode of operation, the CAP input (selected by the CTCR bits 3:2) is sampled on every rising edge of the PCLK clock. After comparing two consecutive samples of this CAP input, one of the following four events is recognized: rising edge, falling edge, either of edges or no changes in the level of the selected CAP input. Only if the identified event occurs and the event corresponds to the one selected by bits 1:0 in the CTCR register, will the Timer Counter register be incremented.

Effective processing of the externally supplied clock to the counter has some limitations. Since two successive rising edges of the PCLK clock are used to identify only one edge on the CAP selected input, the frequency of the CAP input can not exceed one quarter of the PCLK clock. Consequently, duration of the high/low levels on the same CAP input in this case can not be shorter than 1/(2 PCLK).

Table 429. Count Control Register (T[0/1/2/3]CTCR - addresses 0x4000 4070, 0x4000 8070, 0x4009 0070, 0x4009 4070) bit description

Bit	Symbol	Value	Description	Reset Value
1:0	Counter/ Timer Mode		This field selects which rising PCLK edges can increment the Timer's Prescale Counter (PC), or clear the PC and increment the Timer Counter (TC).	00
		00	Timer Mode: the TC is incremented when the Prescale Counter matches the Prescale Register. The Prescale Counter is incremented on every rising PCLK edge.	
		01	Counter Mode: TC is incremented on rising edges on the CAP input selected by bits 3:2.	
		10	Counter Mode: TC is incremented on falling edges on the CAP input selected by bits 3:2.	
		11	Counter Mode: TC is incremented on both edges on the CAP input selected by bits 3:2.	

Table 429. Count Control Register (T[0/1/2/3]CTCR - addresses 0x4000 4070, 0x4000 8070, 0x4009 0070, 0x4009 4070) bit description ...continued

Bit	Symbol	Value	Description	Reset Value
3:2	Count Input Select		When bits 1:0 in this register are not 00, these bits select which CAP pin is sampled for clocking.	00
		00	CAPn.0 for TIMERN	
		01	CAPn.1 for TIMERN	
		10	Reserved	
		11	Reserved	
		Note: If Counter mode is selected for a particular CAPn input in the TnCTCR, the 3 bits for that input in the Capture Control Register (TnCCR) must be programmed as 000. However, capture and/or interrupt can be selected for the other 3 CAPn inputs in the same timer.		
31:4	-	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

21.6.4 Timer Counter registers (T0TC - T3TC, 0x4000 4008, 0x4000 8008, 0x4009 0008, 0x4009 4008)

The 32-bit Timer Counter register is incremented when the prescale counter reaches its terminal count. Unless it is reset before reaching its upper limit, the Timer Counter will count up through the value 0xFFFF FFFF and then wrap back to the value 0x0000 0000. This event does not cause an interrupt, but a match register can be used to detect an overflow if needed.

21.6.5 Prescale register (T0PR - T3PR, 0x4000 400C, 0x4000 800C, 0x4009 000C, 0x4009 400C)

The 32-bit Prescale register specifies the maximum value for the Prescale Counter.

21.6.6 Prescale Counter register (T0PC - T3PC, 0x4000 4010, 0x4000 8010, 0x4009 0010, 0x4009 4010)

The 32-bit Prescale Counter controls division of PCLK by some constant value before it is applied to the Timer Counter. This allows control of the relationship of the resolution of the timer versus the maximum time before the timer overflows. The Prescale Counter is incremented on every PCLK. When it reaches the value stored in the Prescale register, the Timer Counter is incremented and the Prescale Counter is reset on the next PCLK. This causes the Timer Counter to increment on every PCLK when PR = 0, every 2 pclk when PR = 1, etc.

21.6.7 Match Registers (MR0 - MR3)

The Match register values are continuously compared to the Timer Counter value. When the two values are equal, actions can be triggered automatically. The action possibilities are to generate an interrupt, reset the Timer Counter, or stop the timer. Actions are controlled by the settings in the MCR register.

21.6.8 Match Control Register (T[0/1/2/3]MCR - 0x4000 4014, 0x4000 8014, 0x4009 0014, 0x4009 4014)

The Match Control Register is used to control what operations are performed when one of the Match Registers matches the Timer Counter. The function of each of the bits is shown in [Table 430](#).

Table 430. Match Control Register (T[0/1/2/3]MCR - addresses 0x4000 4014, 0x4000 8014, 0x4009 0014, 0x4009 4014) bit description

Bit	Symbol	Value	Description	Reset Value
0	MR0I	1	Interrupt on MR0: an interrupt is generated when MR0 matches the value in the TC.	0
		0	This interrupt is disabled	
1	MR0R	1	Reset on MR0: the TC will be reset if MR0 matches it.	0
		0	Feature disabled.	
2	MR0S	1	Stop on MR0: the TC and PC will be stopped and TCR[0] will be set to 0 if MR0 matches the TC.	0
		0	Feature disabled.	
3	MR1I	1	Interrupt on MR1: an interrupt is generated when MR1 matches the value in the TC.	0
		0	This interrupt is disabled	
4	MR1R	1	Reset on MR1: the TC will be reset if MR1 matches it.	0
		0	Feature disabled.	
5	MR1S	1	Stop on MR1: the TC and PC will be stopped and TCR[0] will be set to 0 if MR1 matches the TC.	0
		0	Feature disabled.	
6	MR2I	1	Interrupt on MR2: an interrupt is generated when MR2 matches the value in the TC.	0
		0	This interrupt is disabled	
7	MR2R	1	Reset on MR2: the TC will be reset if MR2 matches it.	0
		0	Feature disabled.	
8	MR2S	1	Stop on MR2: the TC and PC will be stopped and TCR[0] will be set to 0 if MR2 matches the TC.	0
		0	Feature disabled.	
9	MR3I	1	Interrupt on MR3: an interrupt is generated when MR3 matches the value in the TC.	0
		0	This interrupt is disabled	
10	MR3R	1	Reset on MR3: the TC will be reset if MR3 matches it.	0
		0	Feature disabled.	
11	MR3S	1	Stop on MR3: the TC and PC will be stopped and TCR[0] will be set to 0 if MR3 matches the TC.	0
		0	Feature disabled.	
31:12	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

21.6.9 Capture Registers (CR0 - CR1)

Each Capture register is associated with a device pin and may be loaded with the Timer Counter value when a specified event occurs on that pin. The settings in the Capture Control Register determine whether the capture function is enabled, and whether a capture event happens on the rising edge of the associated pin, the falling edge, or on both edges.

21.6.10 Capture Control Register (T[0/1/2/3]CCR - 0x4000 4028, 0x4000 8028, 0x4009 0028, 0x4009 4028)

The Capture Control Register is used to control whether one of the four Capture Registers is loaded with the value in the Timer Counter when the capture event occurs, and whether an interrupt is generated by the capture event. Setting both the rising and falling bits at the same time is a valid configuration, resulting in a capture event for both edges. In the description below, "n" represents the Timer number, 0 or 1.

Note: If Counter mode is selected for a particular CAP input in the CTCR, the 3 bits for that input in this register should be programmed as 000, but capture and/or interrupt can be selected for the other 3 CAP inputs.

Table 431. Capture Control Register (T[0/1/2/3]CCR - addresses 0x4000 4028, 0x4000 8020, 0x4009 0028, 0x4009 4028) bit description

Bit	Symbol	Value	Description	Reset Value
0	CAPORE	1	Capture on CAPn.0 rising edge: a sequence of 0 then 1 on CAPn.0 will cause CR0 to be loaded with the contents of TC.	0
		0	This feature is disabled.	
1	CAP0FE	1	Capture on CAPn.0 falling edge: a sequence of 1 then 0 on CAPn.0 will cause CR0 to be loaded with the contents of TC.	0
		0	This feature is disabled.	
2	CAP0I	1	Interrupt on CAPn.0 event: a CR0 load due to a CAPn.0 event will generate an interrupt.	0
		0	This feature is disabled.	
3	CAP1RE	1	Capture on CAPn.1 rising edge: a sequence of 0 then 1 on CAPn.1 will cause CR1 to be loaded with the contents of TC.	0
		0	This feature is disabled.	
4	CAP1FE	1	Capture on CAPn.1 falling edge: a sequence of 1 then 0 on CAPn.1 will cause CR1 to be loaded with the contents of TC.	0
		0	This feature is disabled.	
5	CAP1I	1	Interrupt on CAPn.1 event: a CR1 load due to a CAPn.1 event will generate an interrupt.	0
		0	This feature is disabled.	
31:6	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

21.6.11 External Match Register (T[0/1/2/3]EMR - 0x4000 403C, 0x4000 803C, 0x4009 003C, 0x4009 403C)

The External Match Register provides both control and status of the external match pins. In the descriptions below, "n" represents the Timer number, 0 or 1, and "m" represent a Match number, 0 through 3.

Match events for Match 0 and Match 1 in each timer can cause a DMA request, see [Section 21.6.12](#).

Table 432. External Match Register (T[0/1/2/3]EMR - addresses 0x4000 403C, 0x4000 803C, 0x4009 003C, 0x4009 403C) bit description

Bit	Symbol	Description	Reset Value
0	EM0	External Match 0. When a match occurs between the TC and MR0, this bit can either toggle, go low, go high, or do nothing, depending on bits 5:4 of this register. This bit can be driven onto a MATn.0 pin, in a positive-logic manner (0 = low, 1 = high).	0
1	EM1	External Match 1. When a match occurs between the TC and MR1, this bit can either toggle, go low, go high, or do nothing, depending on bits 7:6 of this register. This bit can be driven onto a MATn.1 pin, in a positive-logic manner (0 = low, 1 = high).	0
2	EM2	External Match 2. When a match occurs between the TC and MR2, this bit can either toggle, go low, go high, or do nothing, depending on bits 9:8 of this register. This bit can be driven onto a MATn.2 pin, in a positive-logic manner (0 = low, 1 = high).	0
3	EM3	External Match 3. When a match occurs between the TC and MR3, this bit can either toggle, go low, go high, or do nothing, depending on bits 11:10 of this register. This bit can be driven onto a MATn.3 pin, in a positive-logic manner (0 = low, 1 = high).	0
5:4	EMC0	External Match Control 0. Determines the functionality of External Match 0. Table 433 shows the encoding of these bits.	00
7:6	EMC1	External Match Control 1. Determines the functionality of External Match 1. Table 433 shows the encoding of these bits.	00
9:8	EMC2	External Match Control 2. Determines the functionality of External Match 2. Table 433 shows the encoding of these bits.	00
11:10	EMC3	External Match Control 3. Determines the functionality of External Match 3. Table 433 shows the encoding of these bits.	00
15:12	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

Table 433. External Match Control

EMR[11:10], EMR[9:8], EMR[7:6], or EMR[5:4]	Function
00	Do Nothing.
01	Clear the corresponding External Match bit/output to 0 (MATn.m pin is LOW if pinned out).
10	Set the corresponding External Match bit/output to 1 (MATn.m pin is HIGH if pinned out).
11	Toggle the corresponding External Match bit/output.

21.6.12 DMA operation

DMA requests are generated by a match of the Timer Counter (TC) register value to either Match Register 0 (MR0) or Match Register 1 (MR1). This is not connected to the operation of the Match outputs controlled by the EMR register. Each match sets a DMA request flag, which is connected to the DMA controller. In order to have an effect, the GPDMA must be configured and the relevant timer DMA request selected as a DMA source via the DMAREQSEL register, see [Section 31.5.15](#).

When a timer is initially set up to generate a DMA request, the request may already be asserted before a match condition occurs. An initial DMA request may be avoided by having software write a one to the interrupt flag location, as if clearing a timer interrupt. See [Section 21.6.1](#). A DMA request will be cleared automatically when it is acted upon by

the GPDMA controller.

Remark: Because timer DMA requests are generated whenever the timer value is equal to the related Match Register value, DMA requests are always generated when the timer is running, unless the Match Register value is higher than the upper count limit of the timer. It is important not to select and enable timer DMA requests in the GPDMA block unless the timer is correctly configured to generate valid DMA requests.

21.7 Example timer operation

[Figure 115](#) shows a timer configured to reset the count and generate an interrupt on match. The prescaler is set to 2 and the match register set to 6. At the end of the timer cycle where the match occurs, the timer count is reset. This gives a full length cycle to the match value. The interrupt indicating that a match occurred is generated in the next clock after the timer reached the match value.

[Figure 116](#) shows a timer configured to stop and generate an interrupt on match. The prescaler is again set to 2 and the match register set to 6. In the next clock after the timer reaches the match value, the timer enable bit in TCR is cleared, and the interrupt indicating that a match occurred is generated.

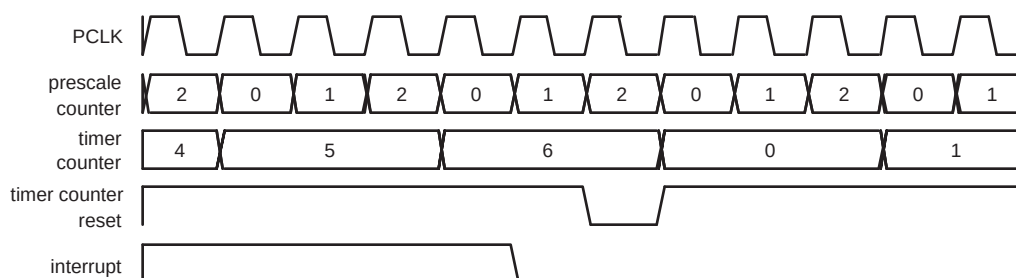


Fig 115. A timer cycle in which PR=2, MRx=6, and both interrupt and reset on match are enabled.

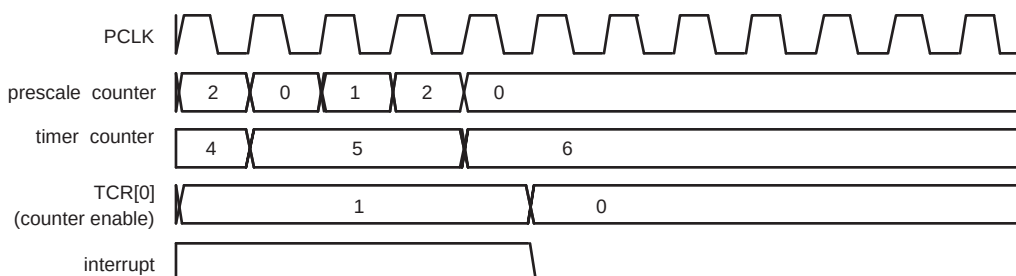


Fig 116. A timer Cycle in Which PR=2, MRx=6, and both interrupt and stop on match are enabled

21.8 Architecture

The block diagram for TIMER/COUNTER0 and TIMER/COUNTER1 is shown in [Figure 117](#).

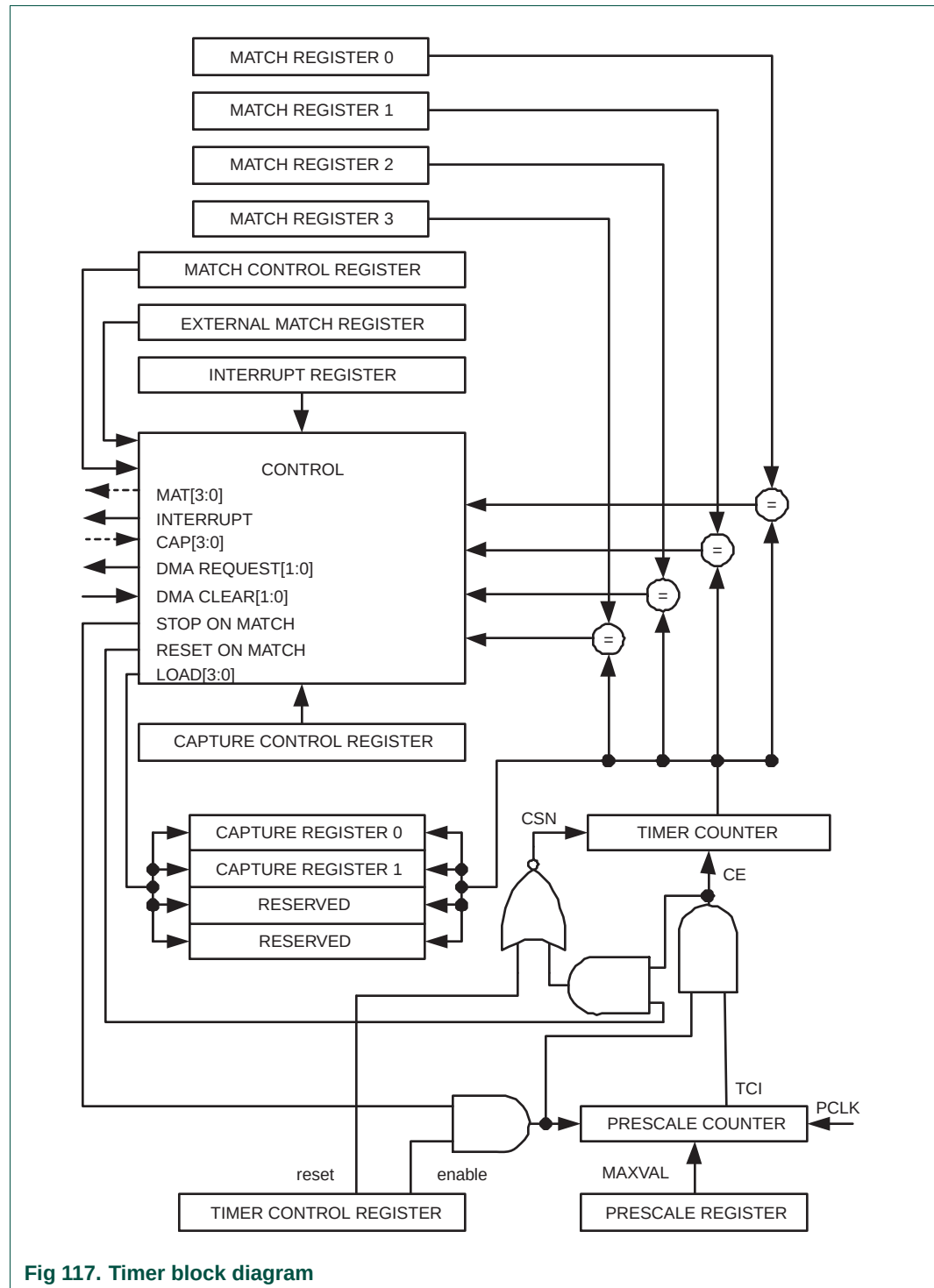


Fig 117. Timer block diagram